Final AI Case Study

Salary Benchmarking Assistant

REFERENCE: V1.0

Track Versions & Modifications

Version	Date	Author	Comments
1.0	01/09/26	Dhanveer Sharma EMRITH	

1 GOAL

Build a small end-to-end pipeline with three independent tasks that each answer "is this salary fair?" in a different way, then an LLM agent that ties them together.

2 DATASET

Path: G:\Shared drives\104. Knowledge & Training\98. New Bees\2025\AI Bootcamp\Final Case Study\2 - Case Study 2026\dataset

Columns: work_year, experience_level (EN/MI/SE/EX), employment_type (FT/PT/CT/FL), job_title, salary, salary_currency, salary_in_usd, employee_residence, remote_ratio (0/50/100), company_location, company_size (S/M/L).

3 TASKS

3.1 TASK 1

This task focuses on comparing each salary against others in a similar role and experience level.

- Group records into peer cohorts (e.g. job_title + experience_level).
- For each cohort, compute the median and the 10th/90th percentile of salary_in_usd.
- Flag any individual salary that falls outside the 10th–90th percentile band of their cohort.
- Output per row: cohort_median, pct_diff_from_median, is_outlier (bool).

Hint: start with job_title + experience_level as the cohort key. By the end, grouping on 4 columns total is the right call; figure out which other 2.

Deliverable: a CSV of the flagged outliers.

3.2 TASK 2

This task looks at what a fair salary should look like for each employee, based on their profile.

- Using experience_level, job_title, remote_ratio, company_size, and company_location, predict the salary for each row of the entire dataset
- Compute the residual salary (actual salary minus predicted salary).

Deliverable: for each row, the predicted salary and the residual.

3.3 TASK 3

This task brings the previous two tasks together behind a single assistant that can answer questions on demand.

- Build an agent with 2 tools: one that returns cohort statistics, one that returns a predicted salary.
- Wire these tools up to an LLM (connection details provided separately) so it can call them to answer questions.

Ask the agent:

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"Is a Head of Data at Executive level in the US, fully on-site, at a "
"medium-sized company, earning $1,000 fairly paid?"
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The agent should decide which tool(s) to call, read the real numbers back, and write its answer grounded in those numbers, not guess or make up a figure itself.

Deliverable: console output showing which tool(s) got called and the agent's final grounded answer.